StudySapiens Complete Chapter Notes

Chapter 1 — Biodiversity and Classification

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter is organised exactly on the textbook's own topics (1.1 to 1.8). Every topic has been converted into the long questions a board paper can ask from it, and each topic ends with conceptual short questions (why / how / differentiate / justify) that are important for the board exam; direct definitions are omitted. The terminology, sequence, examples and data are kept as given in the textbook.

Board pattern (Punjab 2026, new syllabus): Chapter 1 carries 2 MCQs, 2 short questions in Q.2, and part (a) of a long question in Q.5. Long questions are of 8 marks (often split 5+3 or 6+2), short questions of 2 marks each.

TOPIC 1.1 — THREE-DOMAIN SYSTEM OF CLASSIFICATION

Textbook topic 1.1 · printed pages 1–5

Q.1 Describe the three-domain system of classification. How are Archaea unique?

Board long question · 5 + 3 marks

From five kingdoms to three domains

According to the **five-kingdom classification system**, proposed by American ecologist **Robert Whittaker in 1969**, all organisms were divided into five kingdoms i.e., **Monera, Protista, Fungi, Plantae and Animalia**. According to this system, the kingdom Monera included prokaryotes while all the other four kingdoms included eukaryotes.

In **1990**, American microbiologist **Carl Woese** suggested that there are two separate groups of prokaryotes i.e., **Archaea** and **Bacteria**. On this basis, he classified living organisms into **three domains** i.e., domain Archaea, domain Bacteria and domain Eukarya. In this system, domain Archaea and domain Bacteria contain prokaryotes but they differ in a number of features.

Key term

Phylogeny: The evolutionary relationship among organisms is called phylogeny. The diagram to show phylogeny is called a **phylogenetic or evolutionary tree**.

Now biologists believe that Archaea and Bacteria evolved independently from some common ancestor. Molecular evidence suggests that archaea are more closely related to eukaryotes than to bacteria. In other words, **Eukarya evolved from Archaea**, after archaea split off from the Bacteria.

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Domain Archaea

In the five-kingdom system, this domain was included in kingdom Monera. The name Archaea comes from the Greek archaios ("ancient"). They are prokaryotes which diverged from bacteria in very ancient times. Archaea were initially classified as a group of bacteria, and were called **archaebacteria**.

Individual archaeans range from **0.1 µm to over 15 µm** in diameter. Some form aggregates or filaments up to **200 µm** in length. They occur in various shapes, such as spherical, rod-shape, spiral, lobed, or rectangular. Archaea reproduce asexually by **binary or multiple fission, fragmentation, or budding**. Mitosis and meiosis do not occur in archaea.

How are Archaea unique?

1. **Cell Membrane:** Their cell membrane contains lipids with **ether-linkage** between glycerol and fatty acid chains. The fatty acid chains are **branched**. That is why their cell membranes are more resistant to extreme conditions. On the other hand, bacteria and Eukarya have membrane lipids with fatty acids attached to glycerol by **ester linkages**, and the fatty acid chains are **unbranched**.
2. **Cell Wall Composition:** The cell walls of archaea lack cellulose and peptidoglycan. Instead, they contain distinct polysaccharides and proteins. Some archaea have **pseudopeptidoglycan**. Bacterial cell walls contain **peptidoglycan**, a polymer consisting of sugars and amino acids that provides structural support. In Eukarya, the cell walls, if present, are composed of **cellulose** (in plants) or **chitin** (in fungi).
3. **Genetic Differences:** Archaea share several genetic sequences and regulatory features with eukaryotes, highlighting their closer evolutionary relationship.
4. **Metabolism:** Archaea have unique metabolic processes like **methanogenesis** (production of methane), which is not found in bacteria or Eukarya. Bacteria exhibit metabolic pathways including photosynthesis, nitrogen fixation and fermentation. In Eukarya, the metabolic processes are often more complex and include cellular respiration, photosynthesis (in plants and algae), and various forms of fermentation.

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Feature	Archaea	Bacteria	Eukarya
Membrane lipid linkage	Ether linkage; branched chains	Ester linkage; unbranched	Ester linkage; unbranched
Cell wall	No peptidoglycan; polysaccharides + proteins; some pseu dopeptidoglycan	Peptidoglycan	Cellulose (plants) or chitin (fungi); absent in animals
Unique metabolism	Methanogenesis	Photosynthesis, nitrogen fixation, fermentation	Cellular respiration, photosynthesis, fermentation

Significance of Archaea

The archaeans which live in high acidity and alkalinity are a source of **enzymes that can function under harsh conditions**. For example, the enzymes of DNA replication have been extracted from such archaeans. These enzymes can work best at high temperatures and allow rapid cloning of DNA in the laboratory. Similarly, the **methanogen** archaeans are a vital part of **sewage treatment**. They carry out anaerobic digestion and produce **biogas**. **Acidophilic** Archaea are used to extract metals such as gold, cobalt and copper from ores in mineral processing.

Major Groups of Archaea

- **Methanogens** — produce methane as a metabolic by-product.
- **Halobacteria** — live in extremely saline environments.
- **Thermococci** — found in hot environments.
- **Thaumarchaeota** — involved in nitrogen cycling.

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Q.2 Write the general characteristics of the domain Bacteria. Name its major groups.

Board long question · 5 + 3 marks

In the five-kingdom system, this domain was included in kingdom Monera. They are the **true bacteria**. They possess several distinct characteristics that differentiate them from the other domains i.e., Archaea and Eukarya.

1. **Cell Structure:** Like archaea, bacteria possess a **prokaryotic cell** i.e., they lack a true nucleus and membrane-bound organelles.
2. **Cell Wall Composition:** Bacteria have a cell wall composed of **peptidoglycan**, a unique polymer that provides structural support and shape.
3. **Genetic Material:** Like Archaea, bacteria possess a **single, circular chromosome** composed of DNA, located in the **nucleoid** region.
4. **Plasmids:** Most bacteria have small, circular DNA molecules that can be transferred between bacteria, aiding in genetic diversity and adaptation.

5. **Reproduction:** Bacteria primarily reproduce asexually through **binary fission**, a process where a single cell divides into two identical daughter cells.

6. **Nutritional Modes:** Include **autotrophs** (self-feeding, e.g., photosynthetic bacteria) and **heterotrophs** (feeding on organic matter, e.g., decomposers).

7. **Morphology:** Bacteria exhibit a variety of shapes, such as **cocci** (spherical), **bacilli** (rod-shaped), **spirilla** (spiral-shaped), and **vibrios** (comma-shaped).

8. **Arrangement:** Cells may be found singly, in pairs (**diplococci**), chains (**streptococci**), clusters (**staphylococci**), or other arrangements based on species-specific characteristics.

9. **Flagella:** Many bacteria have one or more flagella, whip-like structures that enable movement.

10. **Pili and Fimbriae:** These are hair-like structures in some bacteria. They help in attachment to surfaces and in exchange of genetic material with other bacteria.

11. **Respiration:** Bacteria can be **obligate aerobes, obligate anaerobes, facultative anaerobes, microaerophiles, or aerotolerant anaerobes**. Some bacteria perform fermentation to produce energy in the absence of oxygen.

12. **Extremophiles:** Some bacteria thrive in extreme conditions, such as high temperatures (**thermophiles**), high salinity (**halophiles**), and low pH (**acidophiles**).

13. **Pathogenicity:** Some bacteria cause diseases in humans, animals and plants, producing toxins or other virulence factors.

14. **Symbiosis:** Many bacteria live in symbiotic relationships with other organisms, including **mutualism** (both benefit) and **commensalism** (one benefits, the other is not harmed).

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Major Groups of Bacteria

Group	Examples
Proteobacteria	Escherichia coli, Rhizobium, Helicobacter pylori
Firmicutes	Bacillus subtilis, Lactobacillus, Clostridium botulinum
Actinobacteria	Streptomyces, Mycobacterium tuberculosis
Cyanobacteria	Anabaena, Spirulina
Spirochaetes	Treponema pallidum
Acidobacteria	Acidobacterium
Aquificae	Aquifex pyrophilus

Q.3 Describe the general characteristics of the domain Eukarya that justify its classification as a separate domain.

Board long question · 8 marks

The domain Eukarya encompasses all organisms with **eukaryotic cells**, which are fundamentally different from the prokaryotic cells of Bacteria and Archaea.

1. **Cell Structure:** They possess eukaryotic cells — with a **true nucleus** enclosed by a nuclear membrane. Cells have **membrane-bound organelles** e.g., mitochondria, chloroplasts (in plants and algae), endoplasmic reticulum, Golgi apparatus, lysosomes and peroxisomes. Cells also have a **cytoskeleton** i.e., a complex network of microtubules, microfilaments and intermediate filaments that provides structural support, enables cell movement and facilitates intracellular transport.
2. **Genetic Material:** Their DNA is organised into **multiple linear chromosomes** within the nucleus. DNA is associated with **histone proteins**, which help in the organisation and regulation of genetic material.
3. **Reproduction:** Most eukaryotes undergo **sexual reproduction** involving meiosis and fertilization, leading to genetic diversity. Some eukaryotes can also reproduce asexually through **mitosis**, producing genetically identical offspring.
4. **Complex Cellular Organization:** In multicellular eukaryotes, cells differentiate into specialised types forming **tissues and organs** with specific functions.
5. **Evolutionary Relationships:** Eukaryotes are believed to have originated through **endosymbiosis**, where certain prokaryotic cells (such as mitochondria and chloroplasts) were engulfed by a host cell, leading to a symbiotic relationship.

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Important Conceptual Short Questions — Topic 1.1

S1. Why are Archaea and Bacteria placed in separate domains although both are prokaryotic?

Archaea have ether-linked, branched membrane lipids and cell walls without peptidoglycan, while Bacteria have ester-linked, unbranched lipids and peptidoglycan walls. Molecular evidence also shows Archaea are more closely related to Eukarya.

S2. How does the membrane lipid structure of Archaea help them survive extreme conditions?

The ether linkage between glycerol and branched fatty acid chains makes their cell membranes more resistant to extreme conditions than the ester-linked, unbranched lipids of Bacteria and Eukarya.

S3. Why are Archaea considered more closely related to Eukarya than to Bacteria?

They share several genetic sequences and regulatory features with eukaryotes. Molecular evidence suggests that Eukarya evolved from Archaea after archaea split off from Bacteria.

S4. How does cell wall composition differentiate the three domains?

Bacteria have peptidoglycan; Archaea lack peptidoglycan and may have pseudopeptidoglycan; Eukarya have cellulose (plants) or chitin (fungi) if a wall is present, while animals lack a cell wall.

S5. Why is methanogenesis considered a unique metabolic feature of Archaea?

Methanogenesis (production of methane) is not found in Bacteria or Eukarya. Methanogens carry out anaerobic digestion in sewage treatment and produce biogas.

S6. How do extremophilic Archaea benefit biotechnology and industry?

Their enzymes function under harsh conditions and high temperatures, allowing rapid DNA cloning in laboratories. Acidophilic archaea are used to extract metals such as gold, cobalt and copper from ores.

S7. How do plasmids contribute to bacterial adaptation?

Plasmids are small, circular DNA molecules that can be transferred between bacteria, aiding in genetic diversity and adaptation.

S8. Why can bacteria occupy such diverse habitats?

They show autotrophic and heterotrophic nutrition, varied respiration types, and some are extremophiles (thermophiles, halophiles, acidophiles). Symbiotic relationships such as mutualism and commensalism also help them survive.

S9. Differentiate between obligate anaerobes and facultative anaerobes in bacteria.

Obligate anaerobes live only in the absence of oxygen, while facultative anaerobes can live with or without oxygen. Some bacteria perform fermentation to produce energy when oxygen is absent.

S10. How do pili and fimbriae increase bacterial survival?

They help bacteria attach to surfaces and exchange genetic material with other bacteria, aiding colonisation and genetic diversity.

S11. How does reproduction in Archaea differ from that in Eukarya?

Archaea reproduce asexually by binary or multiple fission, fragmentation or budding, and mitosis and meiosis do not occur. Most eukaryotes undergo sexual reproduction involving meiosis and fertilization, or asexual reproduction by mitosis.

S12. Justify the modification of the five-kingdom system into the three-domain system.

Whittaker's system placed all prokaryotes in Monera. Woese showed two separate prokaryotic groups, Archaea and Bacteria, so organisms were reclassified into three domains: Archaea, Bacteria and Eukarya.

TOPIC 1.2 — TAXONOMIC HIERARCHY

Textbook topic 1.2 · printed pages 5–7

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Q.4 Explain the taxonomic hierarchy and give the classification of human, sparrow and onion.

Board long question · 5 + 3 marks

The classification of living organisms is organised into a **hierarchical system** that allows scientists to categorise and understand the relationships between different forms of life. This system includes several levels, known as **taxa** (singular: **taxon**), each representing a rank in the biological classification system.

> Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species

Taxonomic hierarchy from the highest to the most specific rank

1. **Domain** — the highest level of classification. Currently there are three domains: Archaea, Bacteria and Eukarya.
2. **Kingdom** — one of the highest taxonomic ranks, just below domain. It groups together all forms of life that share fundamental characteristics. In domain Eukarya the kingdoms include Animalia, Plantae, Fungi and Protista.
3. **Phylum** — the next level below kingdom. Organisms within a phylum share a **basic body plan** and significant structural features. In kingdom Animalia, phylum **Chordata** includes all animals with a notochord.
4. **Class** — further divides organisms within a phylum based on more specific common traits. Within Chordata, class **Mammalia** includes all mammals, characterised by hair and mammary glands.
5. **Order** — categorises organisms within a class based on additional shared characteristics and evolutionary history. Within Mammalia, order **Primates** includes humans, monkeys and apes, characterised by large brains and opposable thumbs.
6. **Family** — groups organisms within an order that are even more closely related. Within Primates, family **Hominidae** includes great apes and humans.
7. **Genus** — a more specific rank within a family, grouping species that are very closely related and often visually similar. Within Hominidae, genus **Homo** includes humans and our closest extinct relatives.
8. **Species** — the most specific level, representing a single type of organism. Members of a species can **interbreed and produce fertile offspring**. Within genus Homo, the species **Homo sapiens** refers to modern humans.

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Classification of three representative organisms

Taxonomic Rank	Human	Sparrow	Onion
Domain	Eukarya	Eukarya	Eukarya
Kingdom	Animalia	Animalia	Plantae
Phylum	Chordata	Chordata	Angiosperms
Class	Mammalia	Aves	Monocots
Order	Primates	Passeriformes	Asparagales
Family	Hominidae	Passeridae	Amaryllidaceae
Genus	Homo	Passer	Allium
Species	Homo sapiens	Passer domesticus	Allium cepa

As we move from species towards domain, the number of organisms in each group increases while the number of shared characteristics decreases.

Important Conceptual Short Questions — Topic 1.2

S1. How does taxonomic hierarchy help in understanding relationships among organisms?

It groups organisms into ranked taxa based on shared characteristics. Organisms in lower taxa share more features and are more closely related, while higher taxa include more diverse organisms.

S2. Why does the number of shared characteristics decrease as we move from species to domain?

Higher ranks include a larger number of organisms with greater diversity, so only fundamental characteristics are shared. Lower ranks contain closely related organisms with many precise common attributes.

S3. How does the classification of human and sparrow show both similarity and divergence?

Both belong to Eukarya, Animalia and Chordata, showing a common basic body plan. They diverge at class level: human is Mammalia, while sparrow is Aves.

S4. Why is species considered the most specific unit of classification?

It represents a single type of organism whose members can interbreed and produce fertile offspring, distinguishing them from individuals of other species.

S5. Justify the placement of humans in order Primates.

Humans share the primate characteristics of large brains and opposable thumbs, along with additional shared characteristics and evolutionary history.

S6. How does the position of onion in the hierarchy differ from that of human?

Onion belongs to Plantae, Angiosperms and Monocots, while human belongs to Animalia, Chordata and Mammalia. This reflects the autotrophic plant body plan versus the heterotrophic animal body plan.

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S7. Why is phylum Chordata defined by the presence of a notochord?

A phylum groups organisms that share a basic body plan and significant structural features; all chordates develop a notochord at some stage of development.

TOPIC 1.3 — SALIENT FEATURES OF KINGDOMS OF DOMAIN EUKARYA

Textbook topic 1.3 · printed pages 7–10

Q.5 Describe the salient features of the kingdoms of domain Eukarya with their major groups and examples.

Board long question · 8 marks

Eukarya consists of kingdoms **Protista, Fungi, Plantae and Animalia**. It includes all eukaryotes which consist of complex, eukaryotic cells containing a nucleus and other membrane-bound organelles.

1. Kingdom Protista

Kingdom Protista includes eukaryotes which are **unicellular or colonial or filamentous or simple multicellular**. Simple multicellular means that they **do not have multicellular sex organs**.

- **Protozoa** — animal-like protists. They are **unicellular and heterotrophic**. Examples: Paramecium, Amoeba, Plasmodium, Trypanosoma.
- **Algae** — plant-like protists. They have **cell walls made of cellulose**, contain **chlorophyll** and are **autotrophs**. Examples: Euglena, diatoms.
- **Myxomycota and Oomycota** — fungi-like protists. They have **hyphae-like structure** and are **saprophytic**, e.g., slime molds and water molds.

> **Important**
>
> Certain protists are parasitic and cause diseases like **malaria** (Plasmodium), **amoebic dysentery** (Entamoeba histolytica), and **sleeping sickness** (Trypanosoma).

2. Kingdom Fungi

Fungi are **eukaryotic, heterotrophic** organisms that are unicellular or multicellular. Their cells are covered by a cell wall made of **chitin** (a polysaccharide). Fungi get nutrients in a unique way — they **do not ingest food** like animals and some protists; they **absorb food from the surroundings**. Examples are mushrooms, rusts, smuts and molds. There are about **100,000 known species** of fungi.

- **Zygomycota** — fungi which **lack septa** in their hyphae. Example: Rhizopus (bread mold), which grows on moist bread and fruits.
- **Ascomycota** — the **largest group** of fungi, with **septate hyphae**. Examples: common molds, morels, truffles, cup fungi, Neurospora and yeasts.

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- **Basidiomycota** — fungi with **septate hyphae**. Examples: mushrooms, toadstools, puffballs, jelly fungi, bracket/shelf fungi, rusts and smuts.

> **Important**
>
> **Penicillin**, an antibiotic, is derived from the fungus Penicillium. Many fungi form **mycorrhizae** (association with plant roots).

3. Kingdom Plantae

It includes plants, which are **eukaryotic, multicellular** organisms with cell walls made of **cellulose**. They are **autotrophic** and prepare food through **photosynthesis**. **All plants develop from embryos.** Examples are mosses, ferns, conifers and flowering plants.

- **Nonvascular plants (bryophytes)** — lack conducting tissues (xylem and phloem). Examples: liverworts, hornworts and mosses.
- **Vascular plants** — have conducting tissues. They are of two types: **seedless plants** (e.g., ferns) and **seed plants** (e.g., conifers and flowering plants).
- **Gymnosperms** — seed-producing vascular plants with **naked seeds**; cone-bearing, with no flowers or fruits (e.g., Pine and Spruce).
- **Angiosperms** — flowering plants, the **most diverse plant group**, which produce seeds **enclosed in fruits** (Monocots e.g., grass, corn; Dicots e.g., rose, sunflower).

4. Kingdom Animalia

This kingdom of eukaryotes includes animals which are **eukaryotic, multicellular and heterotrophic**. They **develop from embryos**. They **ingest food and digest it within their bodies**.

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Q.6 Give the distinguishing characteristics of the kingdoms of the three domains in tabular form.

Board long question · 8 marks (table question)

Character	Bacteria (Monera)	Archaea (Monera)	Eukarya Protista	Eukarya Fungi	Eukarya Plantae / Animalia
Cell type	Prokaryotic	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Nuclear envelope	Absent	Absent	Present	Present	Present
Presence of cell wall	In all	In all	In some	In all	Plantae: in all Animalia: absent
Composition of cell wall	Peptidoglycan	Various chemicals	Various chemicals	Chitin	Plantae: cellulose and other polysaccharides Animalia: no cell wall
Mode of nutrition	Autotroph or heterotroph	Autotroph or heterotroph	Photosynthetic or heterotroph, or combination	Absorptive heterotroph	Plantae: photosynthetic autotrophs Animalia: ingestive heterotroph
Multicellularity	Absent	Absent	Absent in most forms	Present in most forms	Present in all forms

This is the textbook table "Distinguishing Characteristics of the kingdoms of three domains". Reproduce it as a table in the board exam — tables carry full marks and are faster to write than paragraphs.

Important Conceptual Short Questions — Topic 1.3

S1. Why are fungi not included in Kingdom Plantae despite having cell walls?

Fungal walls are made of chitin, not cellulose, and fungi are absorptive heterotrophs. Plants are photosynthetic autotrophs with cellulose walls.

S2. How does the mode of nutrition differentiate Fungi from Animalia?

Fungi absorb food from the surroundings, while animals ingest food and digest it within their bodies.

S3. Why are protists described as simple multicellular?

Simple multicellular means that they do not have multicellular sex organs, even if they are colonial, filamentous or multicellular.

S4. How do algae differ from fungi-like protists?

Algae are plant-like, autotrophic protists with cellulose walls and chlorophyll. Fungi-like protists (Myxomycota and Oomycota) have hyphae-like structures and are saprophytic.

S5. Why are Zygomycota separated from Ascomycota and Basidiomycota?

Zygomycota lack septa in their hyphae, whereas Ascomycota and Basidiomycota have septate hyphae.

S6. How does the presence of vascular tissue differentiate bryophytes from ferns and seed plants?

Bryophytes lack conducting tissues (xylem and phloem), while ferns and seed plants are vascular and possess these conducting tissues.

S7. Why are gymnosperms called naked-seeded plants?

They produce seeds that are not enclosed in fruits; they are cone-bearing and have no flowers or fruits.

S8. How do angiosperms differ from gymnosperms in seed protection?

Angiosperms produce seeds enclosed in fruits, while gymnosperms have naked seeds. Angiosperms are also the most diverse plant group.

S9. How does cell wall composition help distinguish the kingdoms of Eukarya?

Protista may have various chemicals, Fungi have chitin, Plantae have cellulose and other polysaccharides, and Animalia lack a cell wall.

S10. Why are certain protists medically important?

Some are parasitic and cause diseases such as malaria (Plasmodium), amoebic dysentery (Entamoeba histolytica) and sleeping sickness (Trypanosoma).

TOPIC 1.4 — CLASSIFICATION OF KINGDOM ANIMALIA

Textbook topic 1.4 · printed pages 10–19

The kingdom Animalia is broadly divided into the phyla studied below. Each phylum is a separate board long question, because the board asks "Describe the structure and systems of organisms in Phylum _____".

Q.7 Describe the structure and systems of organisms in Phylum Porifera.

Board long question · 8 marks

This phylum contains **sponges**. Most of them are **marine** while some live in freshwater. Leucosolenia and Euplectella (Venus' flower basket) are marine sponges. Spongilla is a common freshwater sponge.

- **Organization:** Sponges **do not have tissue level organization**. Most sponges are **asymmetrical** but some have **radial symmetry**. They do not have a nervous system.
- **Canal system:** There are numerous pores in the body wall called **ostia**, through which water enters the body. The larger pore through which water leaves the body is the **osculum**.

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- **Body layers:** The outer layer is made of thin, flat cells called **pinacocytes**. The second layer is jelly-like and is called **mesohyle**, containing **amoeboid cells**. The third layer, which lines the **spongocoel**, is made of **choanocytes** or collar cells.
- **Skeleton:** In the form of microscopic needles of **calcium carbonate or silica**.
- **Reproduction:** Most sponges reproduce asexually by **budding or regeneration**. Some form resistant capsules called **gemmules** containing masses of amoeboid cells. When the parent sponge dies it releases its gemmules; in a favourable environment the amoeboid cells come out and form a new sponge.

Important

A commercial sponge is prepared by **drying, beating and washing** a sponge until all cells are removed.

Q.8 Describe the structure and systems of Phylum Cnidaria. How do cnidarians show two body forms?

Board long question · 5 + 3 marks

Almost all cnidarians are **marine**, although a few are found in freshwater e.g., jellyfish and Hydra. Most cnidarians are **colonial** e.g., Obelia, corals and sea fans. Most of them are **sessile** e.g., Hydra, coral, Obelia, while some are **motile** e.g., jellyfish.

- **Symmetry and germ layers:** They are **radially symmetrical** and **diploblastic** i.e., the adult body contains two tissue layers — the **epidermis** and the **gastrodermis**, derived from ectoderm and endoderm respectively. Between them a jelly-like **mesoglea** is present, containing amoeboid cells.
- **Cnidocytes:** They possess special cells called **cnidocytes**. A cnidocyte contains a special organelle called a **nematocyst**, which **defends the body and captures prey**.
- **Digestive system:** Cnidarians have a blind-ending cavity called the **gastrovascular cavity or enteron**. It opens outside by a single opening, the **mouth**, which also acts as the **anus**. This is called a **sac-like digestive system**. The mouth is surrounded by **tentacles**.
- **Nervous system:** A **network of neurons** in the body wall; there is **no central nervous system**.
- **Absent systems:** They do not have respiratory, excretory and transport systems.

Two body forms

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Polyp	Medusa
Shape: cylindrical.	**Shape:** umbrella-like.
Habit: attached to a substrate at the aboral end (sessile).	**Habit:** free-swimming.
Reproduction: asexual.	**Reproduction:** sexual.

Important

Corals are colonial cnidarians. They produce a hard exoskeleton of **calcium carbonate**; this skeleton makes **coral islands and reefs**.

Q.9 Describe the structure and systems of Phylum Platyhelminthes.

Board long question · 8 marks

They are called **flatworms**. They are mostly **unsegmented** and the body is soft and **dorsoventrally compressed**. Most of them are **free-living** e.g., Planaria. Some are **endoparasites** of humans and other animals e.g., liver fluke, tapeworm and blood fluke.

- **Germ layers:** They are **triploblastic** i.e., the tissues of the body are derived from three embryonic layers — ectoderm, mesoderm and endoderm.
- **Body cavity:** They are **acoelomates**. A loose connective tissue called **parenchyma** fills the space between the body wall and body organs.
- **Symmetry: Bilateral symmetry**, with distinct left and right sides as well as dorsal and ventral sides.
- **Excretory system:** A network of tubular **protonephridia**. Each branch ends in a bulb-like **flame cell**. The cilia of flame cells beat to suck surrounding fluid into the tubules; the tubules filter waste and release it through a small opening called a **nephridiopore**.
- **Nervous system:** A network of neurons with **cerebral ganglia** in the anterior end (head), attached to **longitudinal nerve cords** interconnected by transverse branches. Most free-living flatworms have two simple **eyespots** at the anterior end.
- **Absent systems:** They do not have respiratory and circulatory (transport) systems.
- **Reproduction:** Asexually by **fission** — the animal constricts in the middle, divides into two pieces, and each piece **regenerates** the missing part. Sexually reproducing flatworms are **hermaphrodites (bisexual)**.

Tapeworms belonging to phylum Platyhelminthes are segmented.

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Q.10 Describe the structure and systems of Phylum Nematoda. Why are they called pseudocoelomates?

Board long question · 5 + 3 marks

They are **roundworms** with an elongated, worm-like (round) body with **pointed ends**. Some roundworms are **free-living** in water and soil e.g., Caenorhabditis elegans. Many are **parasites** e.g., Ascaris, hookworm, pinworm and whipworm.

- They are **triploblastic, bilaterally symmetrical** and possess an **unsegmented** body.
- **Pseudocoelom:** They are **pseudocoelomates** because they possess a **false body cavity** called a **pseudocoelom** filled with fluid.
- **Digestive system: Tube-like**, consisting of an alimentary canal with **two openings** — mouth at the anterior end and anus at the posterior end. Parasitic roundworms have simplified digestive systems.
- **Excretory system:** Two main types — the **glandular type** with a single **renette cell** leading to a duct and pore, and the **tubular type** with an **H-shaped** structure of ducts.
- **Nervous system:** A network of neurons with a **nerve ring** around the pharynx, attached to **four longitudinal nerve cords**. Raised hair-like sense organs called **sensory papillae** are present on the lips.
- **Absent systems:** They do not have defined respiratory and circulating systems.
- **Reproduction:** They are **unisexual** — males have testes and females have ovaries.

> **Important**
>
> The pseudocoelomates are classified in **seven phyla**, grouped as a unit called **Aschelminths**. Phylum Nematoda is the representative phylum of this group.

Q.11 Describe the structure and systems of Phylum Mollusca.

Board long question · 8 marks

Molluscs have **soft, unsegmented bodies**. They are widely distributed in natural habitats. Some are exclusively **aquatic** e.g., mussels, octopus and oyster; others live in **moist places** e.g., land snail.

- Molluscs are **triploblastic** with **bilateral symmetry** and possess a **true coelom**. Among coelomates they are included in the group called **protostomes**.
- **Body parts:** The body is divided into three parts — **head**, **visceral mass** (contains organs of digestion, excretion and reproduction), and **foot** (attached with the visceral mass).

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- **Mantle:** An epithelial envelope around the visceral mass. The space between the mantle and visceral mass is the **mantle cavity**. In most molluscs the outer surface of the mantle secretes a **calcareous shell**.
- **Radula:** All molluscs (except Bivalvia) have a rasping, tongue-like organ called the **radula**.
- **Circulatory system:** All of them (except cephalopods) have an **open blood circulatory system**. The heart consists of a **single ventricle and two auricles**.
- **Digestive system: Tube-like**, with two openings — mouth and anus.
- **Excretory system:** Paired tubular structures called **nephridia**. Wastes are gathered from sinuses and discharged into the coelom around the heart; the nephridia open into this coelom. Tiny cilia around their openings move fluid from the coelom into the nephridia, which discharge waste into the **mantle cavity**.
- **Respiration: Gills** work for the exchange of gases.
- **Nervous system: Three pairs of interconnected ganglia** present in the head, visceral mass and foot, interconnected by nerve cords.
- **Locomotion and reproduction:** They move with the help of a **muscular foot**. Most are motile while some are sessile. Most molluscs are **unisexual**.

Key term

In an **open-type** circulatory system the blood does not remain in the vessel; rather, it directly bathes the cells in tissue spaces.

Q.12 Explain the structure and systems of organisms in Phylum Annelida.

Board long question · 8 marks (repeatedly asked)

Annelids are commonly called **segmented worms**. They are found in **marine water** (e.g., Nereis), **freshwater** (e.g., leech) and in **damp soil** (e.g., earthworm). Some annelids are **ectoparasites** e.g., leeches.

- **Metameric segmentation:** The body is divided transversely into a number of similar parts called **segments**. Internally the segments are separated by cross walls called **septa**. Each segment has its own circulatory, excretory and neural elements. This type of segmentation is called **metameric segmentation**.
- **General characters:** Annelids are **bilaterally symmetrical** and **triploblastic**. They are **protostome coelomates**.
- **Setae:** Special chitinous bristles called **setae** are present; they are **absent in leeches**. The body wall is surrounded by a moist, acellular **cuticle** secreted by the epidermis.
- **Digestive system: Tube-like**, divided into distinct parts each performing a specific function. Parasitic annelids have a simplified digestive system.

- **Excretory system:** Ciliated, funnel-shaped **metanephridia**; each segment has **one pair**.
- **Circulatory system: Closed type** — blood always flows in blood vessels. They have specialised pulsating blood vessels called **pseudohearts**. Blood of most annelids has the respiratory pigment **haemoglobin dissolved in the blood plasma**.
- **Respiration:** Gaseous exchange occurs **through the skin**.
- **Nervous system:** A **cerebral ganglion or brain** in the anterior segment. A **double, longitudinal ventral nerve cord** arises from the brain and gives nerves in each segment; ganglia are also present in each segment.
- **Receptors:** Tactile receptors, chemoreceptors, balance receptors and photoreceptors. Some annelids also have well-developed **eyes with lenses**.
- **Reproduction:** Most annelids are **hermaphrodite** (e.g., earthworm, leech) and some are **unisexual** (e.g., Nereis).

The word Annelida means "little rings" — the setae are borne on little rings on the body surface.

Q.13 Describe the structure and systems of Phylum Arthropoda. Why are arthropods the most successful invertebrates?

Board long question · 8 marks

Arthropods are the **most successful of all invertebrates**. They are found in **every type of habitat**, and many terrestrial members can also **fly**. They are **triploblastic, bilaterally symmetrical, protostome coelomates**; the coelom is reduced and present only around the reproductive and excretory systems.

- **Jointed appendages:** Modified for specialised functions e.g., running, crawling, swimming, capturing prey, respiration and reproduction. The appendages around the mouth are modified in different ways to form **mouth parts**.
- **Tagmata:** The body is segmented; some segments are fused to form specialised body regions called **tagmata — head, thorax and abdomen**.
- **Exoskeleton and ecdysis:** They have an **exoskeleton or cuticle** secreted by the epidermis, made chiefly of **chitin**. In young arthropods the exoskeleton is shed from time to time; after shedding, the animal grows at a fast rate and then re-secretes a new exoskeleton. This process is called **ecdysis or molting**.
- **Circulatory system: Open type**. Blood flows in a **hemocoel**, derived from an embryonic cavity called the **blastocoel**. Their blood is colourless as it is without haemoglobin, and is known as **haemolymph**.
- **Respiratory system:** Most arthropods have air tubes called **trachea**, which open out through openings called **spiracles**. Aquatic arthropods respire through **gills**.
- **Digestive system: Tube-like**; the alimentary canal is divided into different parts.

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- **Excretory system: Malpighian tubules** — narrow tubules projected from the alimentary canal, attached at the junction of the **midgut and hindgut**. Nitrogenous wastes are excreted as **solid uric acid crystals**.
- **Nervous system:** Well-developed central nervous system with **three fused pairs of cerebral ganglia (brain)** in the head, and a **double ventral nerve cord** with ventral ganglia in each segment. They have well-developed **compound eyes and antennae**.
- **Reproduction:** They are **unisexual**. They can swim, crawl or fly depending on their habitat.

Why most successful: jointed appendages modified for many functions, a protective chitinous exoskeleton, tracheal respiration on land, and the ability to fly together allow arthropods to occupy every type of habitat.

Important arthropods: insects (mosquito, butterfly, moth, wasp, beetles, grasshopper), crabs, lobsters, prawn, shrimps, crayfishes, spider, tick, mite, scorpion, centipedes and millipedes.

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Q.14 Describe the structure and the water-vascular system of Phylum Echinodermata.

Board long question · 5 + 3 marks

They are **exclusively marine** animals. Some are flattened like a biscuit (e.g., cake urchin), some star-shaped with short arms (e.g., sea star or starfish), some globular (e.g., sea urchin), some star-shaped with long arms (e.g., brittle star), and some elongated (e.g., sea cucumber).

- They are **triploblastic** and **deuterostome coelomates**. Their **larvae are bilaterally symmetrical** but the **adults show radial symmetry** — body parts arranged in **five, or multiples of five**, around an oral–aboral axis.
- **Endoskeleton:** A **calcareous endoskeleton** in the form of plates called **ossicles**. These plates are derived from mesoderm but also come out of the skin and make **spines** on the skin.
- **Digestive system: Tube-like** — mouth leads to oesophagus, stomach, intestine and rectum; the rectum opens out through the anus.
- **Absent systems:** There are **no specialised organs for respiration and excretion**.
- **Nervous system:** Poorly developed, made of a **nerve net**, a **nerve ring**, and five (or multiple of five) **radial nerves**. Most sensory receptors are distributed over the body surface and tube feet.
- **Reproduction:** Echinoderms are **unisexual**. Asexual reproduction involves division of the body followed by **regeneration** of each half.

Water-vascular system

Echinoderms have a **water-vascular system** consisting of tubes and spaces present in the coelom. A **ring canal** surrounds the mouth. It opens outside through a sieve-like plate called the **madreporite**. Five (or a multiple of five) **radial canals** branch from the ring canal. Many **lateral canals** emerge from each radial canal, and each lateral canal ends at a **tube foot**.

> Madreporite → Ring canal → Radial canals → Lateral canals → Tube feet

Path of water in the water-vascular system

Tube feet are extensions of the water-vascular system. They **extend and attach** to a substrate; when water is drawn back from the attached tube feet, they **contract** and the body is pulled forward.

> **Important**
>
> Echinoderms are able to **regenerate lost parts**. Some, especially sea stars and brittle stars, drop various parts when under attack and then regenerate them.

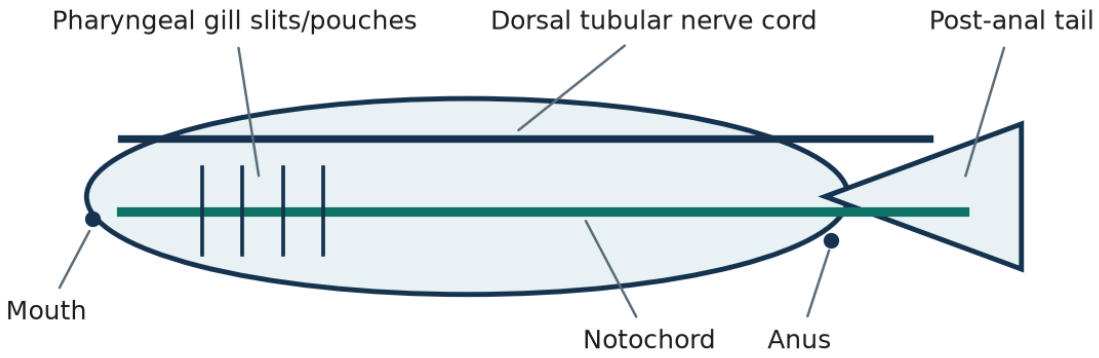
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Q.15 Describe the four unique characteristics of Phylum Chordata and its major groups.

Board long question · 5 + 3 marks (diagram question)

Chordates are **bilaterally symmetrical, triploblastic, deuterostome coelomates**. The following four characteristics are **unique to chordates**, present at some stage in development.

1. **Notochord:** All chordates develop a notochord during embryonic life. It is a **rod-like, semi-rigid body of vacuolated cells** that extends throughout the length of the body **between the gut and the dorsal nerve cord**. Lower chordates retain the notochord throughout life, while in vertebrates it is **partly or entirely replaced by the vertebral column** during development.
2. **Pharyngeal slits:** A series of openings in the lateral walls of the pharynx. All chordates develop **paired gill slits** in the embryonic stage. In some (e.g., Amphioxus and fishes) these develop into **gills**; in most amphibians they are functional for some period of life history; in reptiles, birds and mammals they are **modified for various purposes**.
3. **Tubular nerve cord:** In all chordates a tubular nerve cord runs through the longitudinal axis of the body, **just dorsal to the notochord**. It expands anteriorly as a **brain**.
4. **Post-anal tail:** All chordates develop a tail **posteriorly beyond the anal opening**. Some chordates retain it throughout life while others degenerate it during embryonic life.



Diagnostic characters of chordates — notochord, dorsal tubular nerve cord, pharyngeal gill slits and post-anal tail.

Major groups of Phylum Chordata

- **Subphylum Urochordata** — invertebrate chordates in which the notochord and nerve cord are present **only in their free-swimming larvae**. Example: **sea squirts**.
- **Subphylum Cephalochordata** — invertebrate chordates in which the notochord **persists throughout life**. Example: Amphioxus.
- **Vertebrates** — they have a **vertebral column and cranium**. Vertebrates are divided into seven classes which are placed into two groups.

Important Conceptual Short Questions — Topic 1.4

S1. How does the canal system of sponges compensate for the absence of transport and digestive systems?

Water enters through ostia and leaves through the osculum, bringing food and oxygen and removing wastes. Choanocytes line the spongocoel and help maintain this water current.

S2. Why are sponges said to have cellular level of organization?

They do not have tissue level organization; their body is made of specialized cells such as pinacocytes, amoeboid cells and choanocytes rather than true tissues.

S3. How do gemmules help sponges survive unfavorable conditions?

Gemmules are resistant capsules containing masses of amoeboid cells. When the parent sponge dies it releases them, and in a favourable environment the cells form a new sponge.

S4. How do nematocysts help cnidarians in feeding and defence?

Nematocysts, present inside cnidocytes, defend the body and capture prey.

S5. Why is the digestive system of cnidarians called sac-like?

The gastrovascular cavity has a single opening, the mouth, which also acts as the anus for the removal of undigested material.

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S6. How do polyp and medusa forms differ in mode of life and reproduction?

Polyps are cylindrical, attached to a substrate at the aboral end and reproduce asexually. Medusae are umbrella-like, free-swimming and reproduce sexually.

S7. How do corals contribute to the formation of islands and reefs?

Corals are colonial cnidarians that produce a hard exoskeleton of calcium carbonate; the accumulated skeletons make coral islands and reefs.

S8. How do flame cells help in excretion in flatworms?

The cilia of flame cells beat to suck surrounding fluid into protonephridial tubules; the tubules filter waste materials and release them through nephridiopores.

S9. Why are flatworms called acoelomates?

They lack a body cavity; a loose connective tissue called parenchyma fills the space between the body wall and body organs.

S10. How does regeneration help flatworms in asexual reproduction?

The animal constricts in the middle and divides into two pieces by fission; each piece then regenerates the missing part, forming complete individuals.

S11. Why are nematodes called pseudocoelomates?

They possess a false body cavity called a pseudocoelom, filled with fluid, instead of a true coelom.

S12. How does the digestive system of nematodes differ from that of cnidarians?

Nematodes have a tube-like alimentary canal with two openings (mouth and anus), while cnidarians have a sac-like cavity with a single opening.

S13. How do parasitic roundworms differ from free-living forms in digestive system?

Parasitic roundworms have simplified digestive systems compared with free-living forms.

S14. How does the mantle protect molluscs?

The mantle is an epithelial envelope around the visceral mass; in most molluscs its outer surface secretes a calcareous shell.

S15. Why is the circulatory system of most molluscs called open?

Blood does not remain in vessels but directly bathes cells in tissue spaces. Cephalopods are the exception.

S16. How does the radula help molluscs in feeding?

The radula is a rasping, tongue-like organ used in feeding. It is present in all molluscs except Bivalvia.

S17. How does metameric segmentation benefit annelids?

Each segment has its own circulatory, excretory and neural elements, allowing efficient localized functioning of the body.

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S18. Why is the circulatory system of annelids more efficient than that of arthropods?

Annelids have a closed system in which blood always flows in vessels, aided by pseudohearts; arthropods have an open system in which haemolymph bathes tissues directly.

S19. How do setae help annelids in locomotion?

Setae are chitinous bristles that provide grip during movement; they are absent in leeches.

S20. Why is the moist skin of earthworms important for respiration?

Gaseous exchange occurs through the skin, which is kept moist by an acellular cuticle secreted by the epidermis.

S21. Why are arthropods considered the most successful invertebrates?

They occur in every habitat, have jointed appendages modified for many functions, a protective chitinous exoskeleton, tracheal respiration, and many terrestrial forms can fly.

S22. How does ecdysis help arthropods grow?

The chitinous exoskeleton is shed periodically; after shedding, the animal grows at a fast rate and then re-secretes a new exoskeleton.

S23. How do Malpighian tubules help arthropods conserve water?

They excrete nitrogenous wastes in the form of solid uric acid crystals, which reduces water loss.

S24. Why is arthropod blood called haemolymph and why is it colourless?

It flows in the hemocoel in an open circulatory system and is without haemoglobin, so it is colourless.

S25. How does the water-vascular system help echinoderms move?

Tube feet extend and attach to a substrate; when water is drawn back from the attached tube feet they contract, pulling the body forward.

S26. How does symmetry change during the life of an echinoderm?

Larvae are bilaterally symmetrical, but adults show radial symmetry with body parts arranged in five or multiples of five around an oral-aboral axis.

S27. How does regeneration help echinoderms escape predators?

Some sea stars and brittle stars drop various parts when under attack and then regenerate the lost parts.

S28. How does the fate of the notochord differ in lower chordates and vertebrates?

Lower chordates retain the notochord throughout life, while in vertebrates it is partly or entirely replaced by the vertebral column during development.

S29. How do pharyngeal slits function differently in fishes and in reptiles, birds and mammals?

In Amphioxus and fishes they develop into gills; in reptiles, birds and mammals they are modified for various purposes.

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S30. Why are urochordates still classified as chordates?

Their free-swimming larvae possess a notochord and nerve cord, even though these structures are not retained in the adult.

TOPIC 1.5 — CLASSIFICATION OF VERTEBRATES

Textbook topic 1.5 · printed pages 19–25

Q.16 Describe the characteristics of Group Pisces and give its classification.

Board long question · 8 marks

Members of Group Pisces are **aquatic vertebrates**, living exclusively in freshwater or marine environments.

- **Body shape: Streamlined or fusiform** (spindle-shaped), which helps in reducing water resistance during swimming.
- **Body covering:** Covered with **scales** (dermal in origin) — **placoid, cycloid or ctenoid** scales, depending on the type of fish.
- **Skeleton:** May be **cartilaginous** (sharks and rays) or **bony** (most modern fish).
- **Respiration:** Through **gills**, usually covered by an **operculum** in bony fishes.
- **Circulation: Closed** circulatory system. The heart is **two-chambered** (one auricle and one ventricle). Blood circulation is **single and incomplete**.
- **Locomotion:** With the help of **paired and unpaired fins**, which also provide balance and direction during swimming.
- **Reproduction:** Mostly **oviparous**, but some are **viviparous**. Fertilisation may be **external** (common in bony fishes) or **internal** (common in cartilaginous fishes).
- **Temperature:** Pisces are **ectothermic (cold-blooded)** — body temperature varies with the environment.

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Classes of Pisces — seven classes placed in two groups

Group	Class	Key Characteristics	Examples
Agnathans (Jawless Fishes)	Class Myxini	Mouth with four pairs of tentacles; five to fifteen pairs of pharyngeal slits.	Hagfish
Agnathans (Jawless Fishes)	Class Cephalaspidomorphi	Sucking mouth; seven pairs of pharyngeal slits.	Lamprey
Gnathostomes (Jawed Fishes)	Class Chondrichthyes	Cartilaginous skeleton; tail fin with large upper lobe; lack swim bladder or lungs.	Sharks, skates, ratfish
Gnathostomes (Jawed Fishes)	Class Osteichthyes	Bony skeleton; pneumatic sac functions as lungs or swim bladder.	Lungfish, coelacanths

Q.17 Describe the characteristics of Class Amphibia.

Board long question · 8 marks

- They have a **bony endoskeleton**. Unlike fishes, amphibians have a **neck** — the first vertebra (**cervical vertebra**) moves against the back of the skull and allows the skull to **nod vertically**.
- **Skin: Smooth (without scales) and moist**. It helps in **gas exchange, temperature regulation, and absorption and storage of water**.
- **Heart: Double-circuit** and **three-chambered**, with **two atria and one ventricle**.
- **Respiration:** By **gills in the larval stage** and by **lungs and skin in the adult stage**.
- **Temperature:** They depend on an external heat source, so they are **ectotherms**. They cannot regulate or maintain a constant body temperature, so they are **poikilothermic** animals and **hibernate in winter**.
- **Reproduction:** Amphibians are **unisexual**; fertilization is usually **external**.

Groups of amphibians

- **Tailed amphibians** — salamander, newts and mud puppies.
- **Tail-less amphibians** — frogs and toads.
- **Leg-less amphibians** — caecilians.

Q.18 Explain the amniotic egg and describe the characteristics of Class Reptilia.

Board long question · 3 + 5 marks

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Amniotic egg

Reptiles are the **first animal group that possess amniotic eggs**. Amniotic eggs make protective **extra-embryonic membranes** i.e., **amnion, allantois and chorion**. These membranes **protect the embryo from drying out, nourish it and enable it to develop on land**. The amniotic egg contains a **large amount of yolk**, the primary food supply for the embryo, and abundant **albumin**, which provides additional nutrients and water. The egg is covered with a **leathery calcareous shell** which is **partly permeable to gases but not to water**.

Characteristics of Reptilia

- **Skin: Dry and scaly**.
- **Skeleton:** The bony endoskeleton is **harder than in amphibians**; the skull is longer.
- **Head movement:** The first two cervical vertebrae — **atlas and axis** — allow more movement of the head.
- **Heart:** The **ventricle is incompletely partitioned** into left and right ventricles.
- **Temperature:** Like amphibians, reptiles are **ectothermic** and use an external heat source for thermoregulation. They cannot keep the body temperature constant and are **poikilotherms**.
- **Reproduction:** Fertilization is **internal**; they are **oviparous** (egg-laying).

Present-day reptiles are **lizards, snakes, tuatara and crocodiles**.

Q.19 Which adaptations are required for aerial mode of life in birds? Describe the characteristics of Class Aves.

Board long question · 8 marks (repeatedly asked)

Flight (aerial) adaptations

- **Feathers:** Form the **flight surfaces** that provide **lift** and aid in **steering**; they also **prevent heat and water loss**.
- **Body shape: Streamlined and spindle-shaped**, reducing air resistance.
- **Wings:** The **forelimbs are modified into wings**.
- **Light bones:** Bones are **light due to large air spaces**.
- **Bill:** A lighter sheath called the **bill replaces the teeth**, reducing weight.
- **Keel:** The sternum bears a large bone called the **keel** for the attachment of **flight muscles**.
- **Air sacs:** The bronchi lead to a complex system of **air sacs** that occupy much of the body and even extend into some bones, connected to lungs made of small air tubes called **parabronchi**.
- **Endothermy:** Birds are **endotherms** and **homeotherms**, maintaining a nearly constant core temperature needed for sustained flight.

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Other characteristics of Aves

- **Digestion:** Bills and tongues are modified for a variety of diets. In many birds a diverticulum of the oesophagus, the **crop**, is a storage structure allowing birds to quickly ingest large quantities of food. A region of the stomach, the **gizzard**, has muscular walls to crush food.
- **Heart: Four-chambered**, with complete separation of atria and ventricles.
- **Nervous system:** Much developed; **vision and hearing** are important senses.
- **Respiration:** External nares open into the pharynx through nasal passages; the pharynx leads to the trachea and then bronchi. The organ of voice, the **syrinx**, is situated at the **lower end of the trachea**.
- **Reproduction:** Like reptiles and mammals, birds have **amniotic eggs** with large amounts of yolk and albumin, covered with a leathery shell. Fertilization is **internal** and development is **external** i.e., they are **oviparous**.

Running birds have secondarily lost the power of flight — ostrich, kiwi, rhea, cassowary and emu. **Flying birds** include pigeon, parrot, crow, eagle and robin.

Q.20 Describe the characteristics of Class Mammalia and its three groups.

Board long question · 5 + 3 marks

Mammalia includes the group of vertebrates which are **nourished by milk from the mammary glands** of the mother, and have **hair on their body**.

- **Skin glands** (developed from epidermis): **sebaceous (oil) glands** secrete an oily secretion; **sudoriferous (sweat) glands** release watery secretions used in **evaporative cooling**; **mammary glands** are functional in females.
- **Teeth:** Most mammals have **two sets of teeth** during their lives — **milk teeth and permanent teeth**.
- **Ear:** External ear or **pinna** is present. The middle ear has a chain of **three bones — incus, malleus and stapes**.
- **Temperature:** Mammals are **endothermic and homoeothermic**.
- **Heart: Four-chambered**.
- **Diaphragm:** A muscular diaphragm separates the coelom into **thoracic and abdominal cavities**.
- **Voice:** Well-developed voice apparatus in the form of a **larynx (with vocal cords)** and **epiglottis**.
- **Reproduction:** Fertilization is **internal**.

Three groups of mammals

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Group	Characteristics	Examples
Monotremes	**Egg-laying** mammals; whole development of the embryo proceeds in the egg. They are **oviparous**.	Duckbill platypus, echidna (spiny anteater)
Marsupials	Females have a **pouch (marsupium)** on the abdomen. They give birth to **immature young** which complete development in the pouch.	Opossum, kangaroo
Placental mammals	The **most advanced** mammals. A **placenta** forms between the mother's uterus wall and the foetus body; the foetus is nourished and its wastes removed through it. They are **viviparous**.	Dolphin, rat, monkey, bat, elephant, human

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Important Conceptual Short Questions — Topic 1.5

S1. How does the body shape of fishes reduce water resistance?

The streamlined or fusiform (spindle-shaped) body reduces water resistance during swimming; fins also provide balance and direction.

S2. Why do bony fishes possess an operculum?

The operculum covers the gills, through which respiration takes place.

S3. How does the heart and circulation of fishes differ from that of amphibians?

Fishes have a two-chambered heart and single, incomplete circulation; amphibians have a three-chambered heart and double-circuit circulation.

S4. Why are amphibians dependent on moist skin?

Their smooth, scaleless skin helps in gas exchange, temperature regulation, and absorption and storage of water.

S5. How does the amniotic egg enable reptiles to reproduce on land?

Extra-embryonic membranes (amnion, allantois and chorion) protect the embryo from drying out and nourish it; the leathery calcareous shell is partly permeable to gases but not to water, allowing development on land.

S6. Why are reptiles better adapted to land than amphibians?

They have dry scaly skin, a harder bony endoskeleton, internal fertilization and amniotic eggs, which reduce dependence on water.

S7. How do birds' bones and air sacs reduce body weight for flight?

Bones are light due to large air spaces, and air sacs occupy much of the body and even extend into some bones, reducing weight and aiding respiration.

S8. Why is the keel important for flight in birds?

The keel on the sternum provides attachment for the flight muscles.

S9. How does the crop help birds in feeding?

The crop is a storage structure that allows birds to quickly ingest large quantities of food.

S10. Why are birds and mammals called homeotherms?

They can maintain a nearly constant core temperature; they are endotherms that obtain heat from cellular processes.

S11. How does the diaphragm help mammals?

It is a muscular partition that separates the coelom into thoracic and abdominal cavities.

S12. How do monotremes differ from other mammals in reproduction?

Monotremes are egg-laying (oviparous) mammals, e.g., duckbill platypus and echidna, whereas most mammals give birth to young ones.

S13. How do marsupials differ from placental mammals in development of young?

Marsupials give birth to immature young that complete development in a pouch (marsupium); placental mammals nourish the foetus through a placenta and give birth to more developed young.

S14. Why are placental mammals considered the most advanced mammals?

A placenta forms between the mother's uterus wall and the foetus body, through which the foetus is nourished and its wastes are removed; they are viviparous.

TOPIC 1.6 — CLASSIFICATION OF VIRUSES

Textbook topic 1.6 · printed pages 25–27

Q.21 How are viruses classified? Describe the basis of virus classification with examples.

Board long question · 8 marks

Viruses are **not considered organisms** because they are **acellular** i.e., not made of cells. They lack any of the characteristics of the three domains of life and are **not classified in any domain and kingdom**.

A virus consists of **nucleic acid (DNA or RNA) surrounded by a protein coat**. They **cannot run any metabolism** and depend upon the **host cell** (including plants, animals and bacteria) to replicate and synthesise their proteins.

Viruses are classified based on their **genetic material, replication strategy, morphology, and the hosts they infect**. The classification follows guidelines established by the **International Committee on Taxonomy of Viruses (ICTV)**.

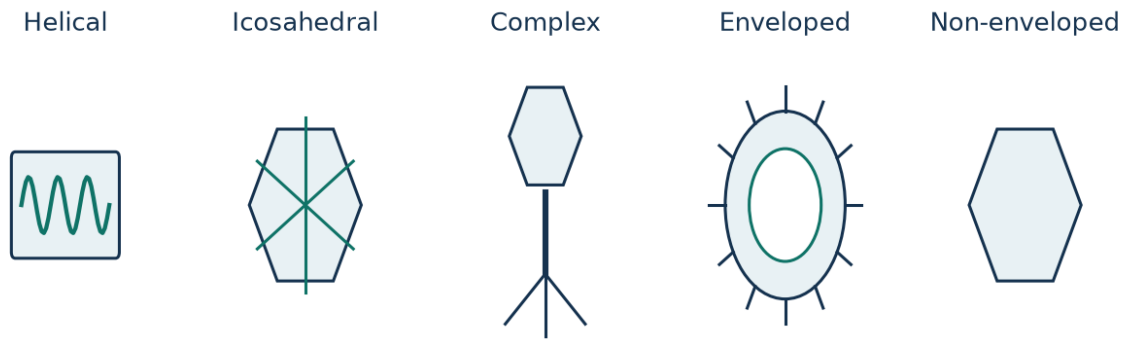
1. Classification on the basis of Host Range

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Type	Host	Examples
Animal viruses	Animals, including humans	Influenza virus, Rabies virus
Plant viruses	Plants	Tobacco mosaic virus, Potato virus X
Bacteriophages	Bacteria	T4 phage, Lambda phage
Archaea viruses	Archaea	Sulfolobus spindle-shaped virus

2. Classification on the basis of Morphology

1. **Helical viruses** — capsid with a **helical structure** surrounding the nucleic acid. Examples: Tobacco mosaic virus, Rabies virus.
2. **Icosahedral viruses** — capsid with a **symmetrical icosahedral shape**. Examples: Adenoviruses, Herpesviruses.
3. **Complex viruses** — a complex structure, often a combination of icosahedral and helical features, sometimes with additional structures like **tails**. Example: Bacteriophages.
4. **Enveloped viruses** — an **outer lipid envelope** derived from the host cell membrane surrounds the capsid. Examples: Influenza virus, HIV.
5. **Non-enveloped (naked) viruses** — **lack an outer lipid envelope**; only a capsid encloses the nucleic acid. Examples: Poliovirus, Adenovirus.



Basic shapes of viruses — helical, icosahedral, complex, enveloped and non-enveloped.

3. Classification on the basis of Genetic Material

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Type	Sub-type	Examples
DNA viruses	dsDNA	Adenoviruses (respiratory infections), Herpesviruses (herpes, chickenpox)
DNA viruses	ssDNA	Parvoviruses (gastroenteritis)
RNA viruses	ssRNA	Coronaviruses (COVID-19), Influenza viruses (flu)
RNA viruses	dsRNA	Rotaviruses (gastroenteritis)

4. Classification on the basis of Replication Strategy

1. **Positive-sense RNA viruses** — the RNA genome is **directly translated** into proteins by the host cell's ribosomes. Examples: Poliovirus, Hepatitis C virus.
2. **Negative-sense RNA viruses** — the RNA genome is **transcribed into mRNA** by a viral RNA polymerase before translation. Examples: Rabies virus, Ebola virus.
3. **Reverse transcribing viruses** — replicate through a **DNA intermediate** using the enzyme **reverse transcriptase**. They may have RNA genomes (Retroviruses like **HIV**, causing AIDS) or DNA genomes (Hepadnaviruses like **Hepatitis B virus**).

Important

Prions and viroids are also acellular and are not considered living organisms. **Prions** are composed of **protein only**; **viroids** are composed of **circular RNA only**. Both cause diseases in certain plants.

Important Conceptual Short Questions — Topic 1.6

S1. Why are viruses not considered living organisms?

They are acellular, cannot run any metabolism, and depend on a host cell to replicate and synthesise their proteins, so they are not classified in any domain or kingdom.

S2. How do viruses replicate without their own metabolic machinery?

They depend upon the host cell (plants, animals or bacteria) to replicate their nucleic acid and synthesise viral proteins.

S3. Why are bacteriophages so named?

They are viruses that infect bacteria, e.g., T4 phage and Lambda phage.

S4. How does an enveloped virus differ from a naked virus?

Enveloped viruses have an outer lipid envelope derived from the host cell membrane surrounding the capsid; naked viruses lack this envelope and consist only of a capsid enclosing the nucleic acid.

S5. How do positive-sense and negative-sense RNA viruses differ in replication?

Positive-sense RNA is directly translated into proteins by host ribosomes; negative-sense RNA must first be transcribed into mRNA by a viral RNA polymerase before translation.

S6. Why are reverse-transcribing viruses unique?

They replicate through a DNA intermediate using the enzyme reverse transcriptase, e.g., HIV (RNA genome) and Hepatitis B virus (DNA genome).

S7. How do prions and viroids differ from viruses?

Prions are composed of protein only and viroids of circular RNA only; both are acellular and cause diseases in certain plants, whereas viruses have nucleic acid surrounded by a protein coat.

TOPIC 1.7 — BIODIVERSITY

Textbook topic 1.7 · printed pages 27–29

Q.22 Explain biodiversity and the levels at which it can be assessed and the importance of random sampling.

Board long question · 5 + 3 marks

Biodiversity, a term derived from "biological diversity", refers to the **variety of life forms present in different ecosystems**, encompassing the diversity of **species, genes and ecosystems**. It represents the **richness and variability of living organisms and their interactions with each other and their environments**.

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Key terms

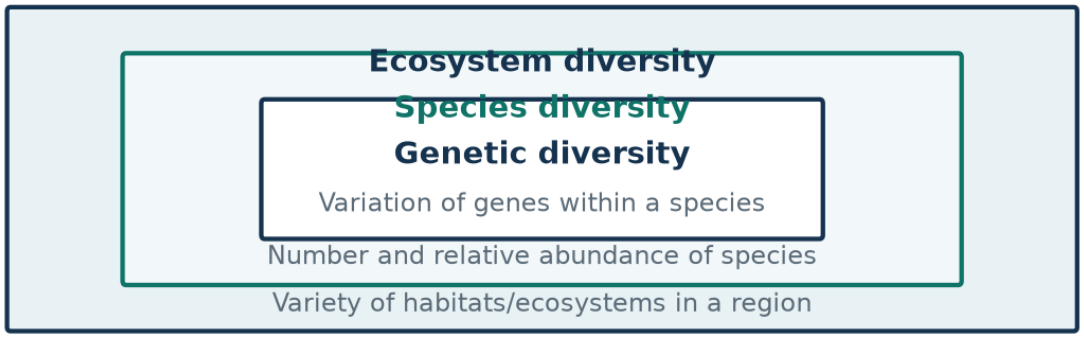
Ecosystem: A dynamic and interactive system composed of living organisms and their physical environment. It includes all the **biotic** as well as the **abiotic** factors.

Niche: The **role or function** of an organism or species within an ecosystem. It includes its habitat, its interactions with other organisms (predation, competition, symbiosis), and its role in energy flow within the ecosystem.

Biodiversity Assessment Levels

1. **Species Level:** Biodiversity is assessed by **identifying and counting the different species** present within a given area. Species diversity includes not only the **number of species** but also their **relative abundance and distribution**.
2. **Genetic Level:** Biodiversity refers to the **variety of genetic information** contained within all individual organisms of a species. This genetic diversity is crucial for the **adaptability and survival of species**, enabling them to cope with environmental changes and challenges.

3. **Ecosystem Level:** Assessment includes the **range of habitats**, from forests and wetlands to grasslands and deserts. It involves understanding how different ecosystems **function** and how they contribute to **overall ecological health**.



The three levels at which biodiversity is assessed.

Importance of Random Sampling in Determining Biodiversity

1. **Minimizes Bias:** It ensures that **every part of the study area has an equal chance** of being sampled, which provides a more accurate representation of the overall biodiversity.
2. **Provides Reliable Estimates:** It allows the collection of data that can be **statistically analysed** to estimate species richness, abundance and distribution.
3. **Facilitates Comparisons:** It enables comparisons between different areas or habitats by providing **standardised methods** of data collection.
4. **Enhances Representativeness:** By covering different parts of the study area, it ensures that the sample **represents the diversity of the entire area**.
5. **Supports Conservation Efforts:** Accurate assessments are essential for **identifying areas of high conservation value** and for **monitoring changes** in biodiversity over time.

Q.23 Describe the methods used to assess the distribution and abundance of organisms in an area.

Board long question · 4 + 4 marks

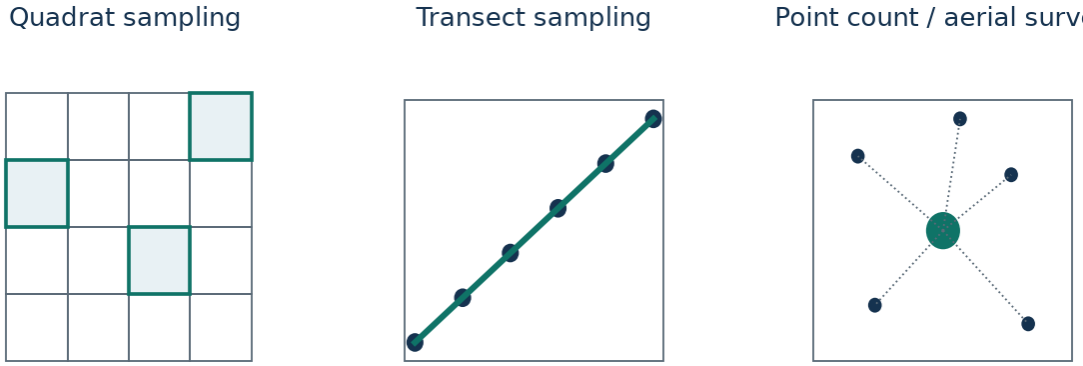
Methods to Assess Distribution

1. **Quadrat Sampling:** Involves **dividing the study area into a grid** and sampling within **randomly selected squares (quadrats)**. It is particularly useful for studying **plant populations or sessile organisms**. For example, in a forest a researcher may lay out quadrats of a fixed size and record the presence or absence of each plant species.

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2. **Transect Sampling:** Involves **laying out a line or strip (transect)** across the study area and **recording species at regular intervals** along this line. It is effective for studying the distribution of species **across environments**. For example, in a coastal zone a transect can be laid from the high tide line to the low tide line to record intertidal organisms.

3. **Aerial Surveys:** Use **aircraft or drones** to observe and record the distribution of organisms over **large areas** — for example, tracking bird species across a large wetland or monitoring large mammal populations in savannas.



Methods to assess the distribution of organisms — quadrat sampling, transect sampling and aerial surveys.

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Methods to Assess Abundance

1. **Point Counts:** Observing and recording the **number of individuals of a species from a fixed point** over a specified period. Commonly used for **birds and other mobile animals**.

2. **Mark-Recapture: Capturing, marking and releasing** individuals of a species, then **recapturing them later** to estimate population size and density. Useful for animals that are **difficult to count directly**.

3. **Quadrat Counts:** Using quadrats to **count the number of individuals** of a species within each quadrat and then inferring these counts to estimate overall abundance.

4. **Capture-Recapture Methods:** Models that account for variables such as **varying capture probabilities and movement between areas**.

5. **Remote Sensing:** Uses **satellite or drone imagery** to assess the abundance and distribution of species, particularly for **large-scale or inaccessible areas**.

Important Conceptual Short Questions — Topic 1.7

S1. Why is random sampling important in biodiversity assessment?

It minimizes bias by giving every part of the study area an equal chance of being sampled, provides reliable estimates, facilitates comparisons, enhances representativeness and supports conservation efforts.

S2. How does genetic diversity help a species survive environmental change?

It provides the variety of genetic information that is crucial for adaptability and survival, enabling species to cope with environmental changes and challenges.

S3. Why is biodiversity assessed at three different levels?

Species, genetic and ecosystem levels each provide unique insights into the richness, adaptability and ecological functioning of life.

S4. How does quadrat sampling differ from transect sampling in application?

Quadrat sampling is useful for plant populations or sessile organisms in randomly selected squares; transect sampling records species along a line to study distribution across environments.

S5. Why is mark-recapture used for mobile animals?

Such animals are difficult to count directly; capturing, marking, releasing and later recapturing them allows estimation of population size and density.

S6. How does remote sensing help in biodiversity studies?

It uses satellite or drone imagery to assess the abundance and distribution of species, particularly for large-scale or inaccessible areas.

S7. Differentiate between habitat and niche.

Habitat is the place where an organism lives; niche is its role or function within an ecosystem, including its interactions with other organisms and its role in energy flow.

TOPIC 1.8 — SPECIES AND SPECIATION

Textbook topic 1.8 · printed pages 29–30

Q.24 Explain the concept of species and speciation. Describe the modes of speciation with examples.

Board long question · 3 + 5 marks

Species

A **species** is generally defined as a **group of individuals that can interbreed and produce fertile offspring under natural conditions**. Members of the same species share common characteristics and genetic makeup, which distinguishes them from individuals of other species.

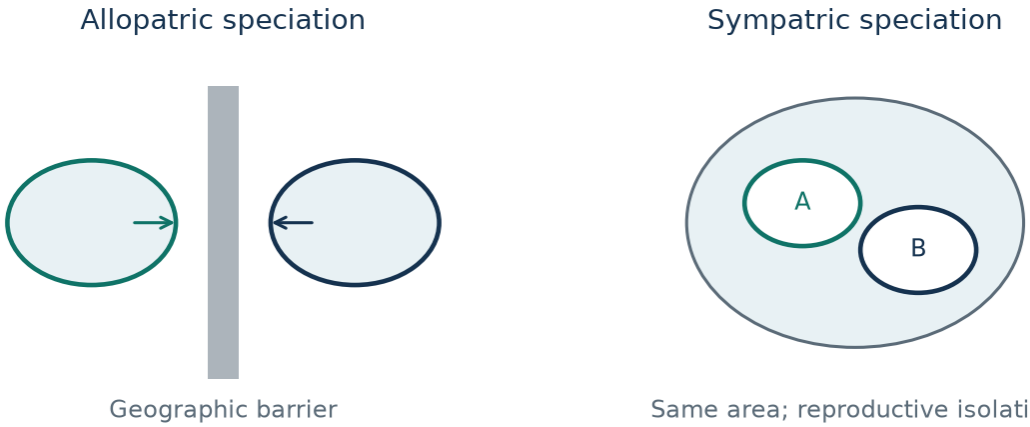
Identification of species by using physical traits and similarities can sometimes be problematic due to the existence of **cryptic species** — organisms that **appear similar but are genetically distinct**. To address this, German-American biologist **Ernst Mayr** emphasised **reproductive isolation** as the key criterion. According to this concept, species are **groups of interbreeding natural populations**; members of different species do not typically mate or produce viable, fertile offspring.

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Speciation

Speciation is the **evolutionary process by which new species arise from a common ancestor**. It involves the **accumulation of genetic changes** that lead to **reproductive isolation** between populations.

1. **Allopatric Speciation:** Occurs when a population is **geographically separated** into two or more isolated groups. These groups experience different environments and **evolve independently**. Over time the accumulated differences can become significant enough to **prevent interbreeding**, even if the geographical barrier is removed. Example: **Darwin's finches** on the Galápagos Islands, where different populations adapted to diverse environments.
2. **Peripatric Speciation:** Involves a **small, isolated population at the edge** of a larger population. The small population undergoes **rapid evolutionary changes**, leading to divergence from the original population. Example: **island species** that evolve from a small founding population.
3. **Parapatric Speciation:** Occurs when populations are **adjacent to each other** but occupy **different environments along a gradient**. **Gene flow is limited** and they evolve adaptations to their specific environments; over time this can lead to reproductive isolation. Example: the grass **Anthoxanthum odoratum**, which exhibits different adaptations to varying soil conditions across a gradient.
4. **Sympatric Speciation:** New species arise **within the same geographical area without physical barriers**. It often occurs through **polyploidy** (where an organism has multiple sets of chromosomes) or **niche differentiation** (where different subpopulations exploit different resources). For instance, certain plants can undergo polyploidy, leading to **immediate reproductive isolation** and the formation of new species.



Allopatric speciation requires a geographic barrier; sympatric speciation occurs in the same area through reproductive isolation.

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Mode	Isolation	Example
Allopatric	Geographical barrier separates populations	Darwin’s finches, Galápagos Islands
Peripatric	Small population isolated at the edge of a larger one	Island species from a small founding population
Parapatric	Adjacent populations along an environmental gradient; limited gene flow	Anthoxanthum odoratum on different soils
Sympatric	Same area, no physical barrier; polyploidy or niche differentiation	Polyploid plants

Important Conceptual Short Questions — Topic 1.8

S1. Why did Ernst Mayr emphasize reproductive isolation over physical similarity?

Cryptic species appear similar but are genetically distinct, so physical traits alone can be misleading. Mayr therefore defined species as groups of interbreeding natural populations.

S2. How does geographical isolation lead to allopatric speciation?

Separated populations experience different environments and evolve independently; over time accumulated differences can prevent interbreeding even if the geographical barrier is removed.

S3. How does peripatric speciation differ from allopatric speciation?

Peripatric speciation involves a small, isolated population at the edge of a larger population that undergoes rapid evolutionary change, rather than large geographically separated groups.

S4. Why can parapatric speciation occur without a complete geographic barrier?

Adjacent populations occupy different environments along a gradient with limited gene flow, allowing them to evolve specific adaptations that can lead to reproductive isolation.

S5. How does polyploidy cause sympatric speciation in plants?

Polyploidy gives an organism multiple sets of chromosomes, leading to immediate reproductive isolation from the parent population within the same geographical area.

S6. Differentiate between allopatric and sympatric speciation.

Allopatric speciation requires a geographical barrier that separates populations; sympatric speciation occurs in the same geographical area without physical barriers, through polyploidy or niche differentiation.

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